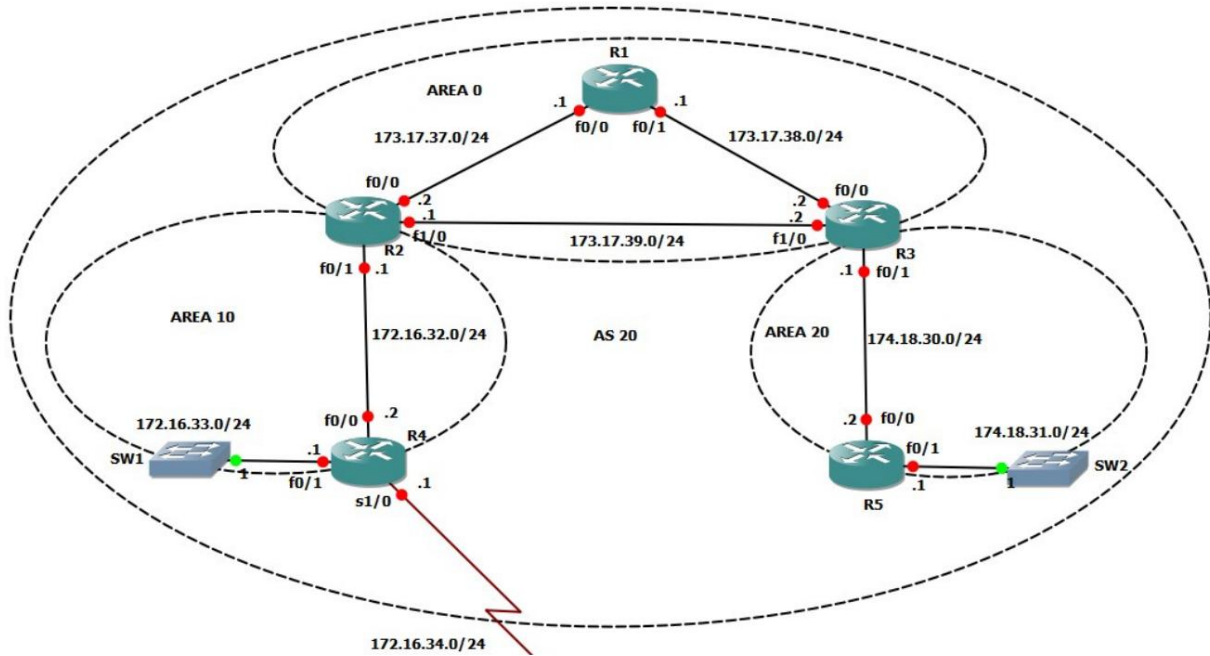




2nd Task: OSPF routing



Assume the above network topology. The topology includes the network number Autonomous System 20. The AS 20 network has been divided into three regions with numbers 0, 10 and 20. To implement point-to-point fast ethernet connections between the routers of the Autonomous System 20, /24 subnets have been assigned to the network administrator.

Create all the connections and assign the corresponding IP addresses to the interfaces.

1. Make the appropriate settings on the routers, so that there is connectivity between all the network devices of the Autonomous System 20. To route traffic between the routers, you should use the OSPF protocol. Specifically, configuring the OSPF protocol on a router involves: § Configuring an OSPF process on the router. § Defining a unique router identifier (optional step). § Defining the subnets that are interconnected on the router and the area to which they belong.
2. Set the f0/1 interfaces of routers R4 and R5 as passive, so that they are not OSPF updates are sent from the specific interfaces.
3. Configure the Area Border Routers of Autonomous System 20 to summarize the routes of the regions that are interconnected (inter-area summarization).
4. Study the neighboring states of routers R1, R2 and R4.



5. Study the OSPF Link State Database of routers R1, R2 and R4 and describe the types of LSAs they contain. 6. Study the routing table of routers R1 and R3 and describe the entries their.

Job deliverables

1. The packet tracer project.
2. Report including answers to questions 1-6

Supporting material

Basic OSPF protocol configuration commands

§ To configure an OSPF routing process, execute the command in global configuration mode: **router ospf <ospf process-id>**. The <ospf process-id> parameter can take any integer value.

§ To define a unique router identifier, run in router configuration mode the command: **router-id <ip-address>**. This command is executed optionally.

§ To define the subnets that are interconnected to a router and the area to which they belong, execute the command in router configuration mode: **network <ip-address> <wildcard-mask> area <area-id>**. The wildcard-mask is calculated from the network mask by replacing each bit 1 with 0 and each bit 0 with 1.

§ To define an interface as passive, execute the command in router configuration mode: **passive-interface <interface_number>**.

§ To summarize the routes of a region advertised by an Area Border Router, run the command in router configuration mode: **area <area-id> range <IP address> <mask>**. The command is repeated for each region that the Area Border Router connects to.

§ To view the router's routing table, execute the command: **show ip route in privileged EXEC mode**.

§ To view information about the enabled routing protocols, execute the command in privileged EXEC mode: **show ip protocols**.

§ To view general information about the OSPF routing process running on a router, run the command in privileged EXEC mode: **show ip ospf <process-id>**.

§ To view the OSPF Link State database, execute the command in privileged EXEC mode: **show ip ospf database**.

§ To view information about a router's neighboring nodes, run privileged EXEC mode the command: **show ip ospf neighbor**.

§ To view OSPF information for an interface, run the following command in privileged EXEC mode:



show ip ospf neighbor <interface_number>.

Route Summary (Route Summarization)

Suppose we want to summarize the following networks. That is, to find an address and a prefix that defines exactly the same range of addresses that define the four networks below:

192.168.0.0 / 24

192.168.1.0 / 24

192.168.2.0 / 24

192.168.3.0 / 24

We convert each IP address to binary.

192.168.0.0	11000000	10101000	000000 00	00000000
192.168.1.0	11000000	10101000	000000 01	00000000
192.168.2.0	11000000	10101000	000000 10	00000000
192.168.3.0	11000000	10101000	000000 11	00000000

The initial prefix of the networks to be summed is 24. We will try to shorten the prefix so that we end up with one network address and not four, while maintaining the total initial address range of the four networks accurately. This range is defined from the first address of the first network to the last address of the last network. For the summation to be possible, two conditions must apply: **a.** the bits of the network portion of the IP address of the summary network, which will be defined by the new smaller prefix, must be identical to the corresponding bits of the networks to be summed

b. The number of bits that are "returned" to the host portion (since we will decrease the prefix, i.e. increase the host portion and decrease the network portion) should be alternated in all possible combinations In our example: • the 22 bits of 192.168.0.0/22 are identical to the first 22 bits of the networks to be

summed. • the 2 bits that will be returned to the host portion are alternated in all combinations 00

01 10 11.

Both conditions are met, so the 4 networks can be combined into one network.

The summarized network mask is derived from the number of common bits of the IP addresses we want to summarize.

To calculate the network id of the summarized network, we set all bits of the host id to 0.

So the IP address of the summarized network is 192.168.0.0/22. **Example of OSPF**

configuration of router R3

Router> **enable**

Router# **configure terminal**



```
Router(config)#router ospf 1
```

```
Router(config-router)#router-id 3.3.3.3
```

```
Router(config-router)#network 173.17.38.0 0.0.0.255 area 0
```

```
Router(config-router)#network 173.17.39.0 0.0.0.255 area 0
```

```
Router(config-router)#network 174.18.30.0 0.0.0.255 area 20
```

```
Router(config-router)#area 20 range 174.18.30.0 255.255.254.0
```

```
Router(config-router)#area 0 range 173.17.36.0 255.255.252.0
```

Websites

[§] IP Routing: OSPF Configuration Guide, Cisco IOS Release 12.4T, <https://goo.gl/jQL484>

§ Cisco OSPF Implementation, <https://goo.gl/khRNlb> § Supporting material in the e-class (exercises, presentations)

Good luck!